



INTRODUCTION TO SOFTWARE ENGINEERING

ASSIGNMENT 02

GROUP MEMEBERS:

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SUBMITTED TO:

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PROJECT OVERVIEW:

This project is a legacy calculator which is written in C++. This calculator is based to find the Loans, To set the payment periods and to get them respectively, to find the total number of payments, and to calculate the loan interest amount with parameters.

The code was changed to fix some issues present in them some are:

1. Bugs where there is integer overflow due to the calculation exceeded the limits of provided float data types.
2. Not provided checks in case if there was an issue where the provided value was not in the range of the required calculations.
3. Not adding standard comments.
4. Another major issue was that there were some header files missing. Additionally, the main function had a pretty complex structure of presenting the cout values.
5. Some other issues included the hardcoded values in the code.

The Technology used:

The tech that we have used for this assignment is:

1. VS code
2. C++ 11
3. Git and GitHub
4. Doxygen
5. CMD

CODE SNIPPETS:

Utils.h and Utils.cpp:

```
string line;
while (getline(file, line))
{
    if (line.empty() || line[0] == '#')
    {
        continue;
    }
    auto pos = line.find('=');
    if (pos == string::npos)
    {
```

```

        continue;
    }
    string key = line.substr(0, pos);
    double value = stod(line.substr(pos + 1));
    config[key] = value;
}
return config;
}

```

This is done to automate the taking of values instead of hardcoding the values

```

double LoanCalculator::calculateLoanBalance()
{
    if (!amountSet_ || !interestSet_ || !periodElapsedSet_ || !paymentSet_)
    {
        throw invalid_argument("Must set loan amount, interest, and elapsed period for this calculation");
    }

    return (amount_ * pow((1 + interestPeriodic_), periodElapsed_) -
            (payment_ / interestPeriodic_) * (pow((1 + interestPeriodic_), periodElapsed_) - 1));
}

```

This is just one example where we have changed nearly all the functions return type and calculations type to double instead of float to fix the issue of integer overflow for large values

```

void setPeriodTotal(int N)
{

```

```
if (N <= 0)
{
    throw std::invalid_argument("Period must be positive");
}

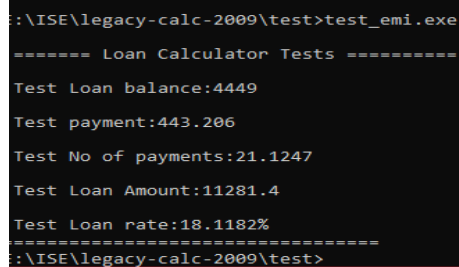
periodTotal_ = N;
periodTotalSet_ = true;
}
```

This again is one such example where we have fixed the issue of keeping check on the values given

Similarly, I have changed the main function to simpler and more easy to read main function.

SCREENSHOTS OF TEST CASES:

There is a bit of issue regarding compiling, libraries so instead of Google Test I have used simple test way in main



```
:\ISE\legacy-calc-2009\test>test_emi.exe
===== Loan Calculator Tests =====
Test Loan balance:4449
Test payment:443.206
Test No of payments:21.1247
Test Loan Amount:11281.4
Test Loan rate:18.1182%
=====
:\ISE\legacy-calc-2009\test>
```

GIT COMMIT HISTORY:

AHMAD 24P-3040

Commits on Nov 25, 2023	
- The final tests are performed in main ● AltimadjaDeveloper committed 5 minutes ago - Doxygen output file added ● AltimadjaDeveloper committed 46 minutes ago - Added comments using Doxygen and made the index.html file ● AltimadjaDeveloper committed 1 hour ago	06df33b <>
Commits on Nov 24, 2023	
- Preformed test_emi.cpp ● AltimadjaDeveloper committed yesterday - refactored code changed the directory to src and fixed the bugs ● AltimadjaDeveloper committed yesterday - Fixed line endigns in window issue ● AltimadjaDeveloper committed yesterday - Updated README.md with build and run instructions ● AltimadjaDeveloper committed yesterday - Add .gitignore ● AltimadjaDeveloper committed yesterday	6f129d8 <>
Commits on Apr 16, 2012	
- Added compilation information ● Brady Johnson committed on Apr 16, 2012 - Nicer formatting on Readme and added better app help text ● Brady Johnson committed on Apr 16, 2012 - Added readme and additional app help text ● Brady Johnson committed on Apr 16, 2012 - Added a Qt Gui and scons compilation. The makefile was generated by Qt. ● Brady Johnson committed on Apr 16, 2012	347e865 <>

```
E:\ISE\legacy-calc-2009>git log --oneline
06df33b (HEAD -> dev, origin/dev) The final tests are performed in main
343edc0 Doxygen output file added
e3222b4 Added comments using Doxygen and made the index.html file
6f129d8 Preformed test_emi.cpp
1e05cd6 refactored code changed the directory to src and fixed the bugs
883d320 Fixed line endigns in window issue
b32bf19 Updated README.md with build and run Instructions
2f386ab Add .gitignore
347e865 (origin/master, origin/HEAD, master) Added compilation information
dff0427 Nicer formatting on Readme and added better app help text
71f0456 Added readme and additional app help text
c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
be10569 Changes to use libCmdLineParser
12ee333 Removed compileCommands.txt
4d09e8b Made methods inline and removed some compiler errors
d99e446 First version
```

FAIZ UL HASSAN 24P-3065

```
C:\Users\Faiz Ul Hassan\Desktop\legacy-calc-2009\legacy-calc-2009>git log --oneline
21ce9e5 (HEAD -> dev-faiz, origin/dev-faiz) feat: complete assignment with personalized contributions
06df33b (origin/dev, dev) The final tests are performed in main
343edc0 Doxygen output file added
e3222b4 Added comments using Doxygen and made the index.html file
6f129d8 Preformed test_emi.cpp
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d99e446 First version
```

```
C:\Users\Faiz Ul Hassan\Desktop\legacy-calc-2009\legacy-calc-2009>
```